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GEOPOLITICS OF RESOURCES

# The Cheapest Monopoly

*The most concentrated mineral supply on earth belongs to a company that has spent seventy years making sure nobody notices.*

2,876 words · 15-minute read

Rank the world's mineral supplies by how much of each comes out of one country and the top of the list is familiar. China mines about 69 per cent of the world's rare earths. The Democratic Republic of the Congo produces about 73 per cent of its cobalt, South Africa about 70 per cent of its platinum. Each of those figures has generated export-control regimes, stockpile legislation, ministerial visits and a substantial library of anxious reports.

Now add the one that beats all three. Brazil mines roughly 93 per cent of the world's niobium, and something between three-quarters and four-fifths of the world total comes from a single privately-held company working a single carbonatite at Araxá, in Minas Gerais [1], [2], [3]. Treating the residual as one competitor — which flatters the tail — the Herfindahl index of niobium mine production comes to about 0.87, against roughly 0.57 for cobalt and for rare earths and 0.58 for platinum. It is the most concentrated significant mineral supply on earth, by a wide margin, and it is held not by a state but by a family holding company.

There has never been a niobium crisis. The element sits on critical-minerals lists, and the United States has been completely import-dependent on it for decades, but nothing resembling the rare-earth alarm of the past fifteen years has ever attached to it. That is the puzzle worth taking seriously, because the answer is not that niobium does not matter. It is in almost every large-diameter gas pipeline, most modern automotive body structures, the magnets of the Large Hadron Collider and the nozzle extensions of rocket engines. The answer is arithmetic, and it runs in two directions at once.

## What Half a Kilogram Does

Niobium goes into steel as ferroniobium, an alloy of roughly 65 per cent niobium in iron, and it goes in at between 0.03 and 0.1 per cent by weight [4]. Take the middle of that band: half a kilogram of niobium in a tonne of steel.

That is a startlingly small quantity to matter. Worked out atom by atom, 0.05 weight per cent niobium in iron is an atomic fraction of 301 parts per million — one niobium atom for every 3,300 atoms of iron. Nothing about the bulk properties of the metal is being altered by dilution at that level. What the niobium does instead is control the size of the crystals, and it does so twice over.

The first mechanism operates while the steel is still being rolled. Niobium in solution, and then as carbides precipitating on the deformed structure, pins the boundaries that austenite grains would otherwise use to recrystallise between rolling passes. Strain therefore accumulates through the finishing mill instead of being annealed out after each bite, and the austenite that finally transforms is heavily worked and finely divided [5]. The second mechanism is what stops the grains growing back, and it has a clean closed form. Smith's 1948 paper on microstructure — which credits the argument to Zener, in a private communication — gives the grain size at which the drag of a dispersion of particles balances the driving force for growth:

$$D_{\text{lim}} = \frac{4r}{3f},$$

where $r$ is the particle radius and $f$ their volume fraction [6]. Gladman refined the criterion in 1966 by accounting for the spread of grain sizes around the mean, which is what makes it usable on a real steel [7].

The volume fraction is not a free parameter; it follows from the addition. Half a kilogram of niobium taken entirely to niobium carbide gives 0.0565 per cent by mass, and at 7.82 against 7.87 grams per cubic centimetre that is 5.68 parts in ten thousand by volume. Strain-induced carbides in austenite are a few nanometres across, so at a radius of 2.5 nanometres those particles sit about 49 nanometres apart and hold the grain size to 5.9 micrometres. Cut the addition to 0.01 per cent and the same particles hold only 29 micrometres, which is to say they hold nothing at all.

What the fine grain is worth comes from the Hall-Petch relation, established by Hall in 1951 and Petch in 1953 [8], [9]:

$$\sigma_y = \sigma_0 + k\,d^{-1/2}.$$

The constant $k$ for ferrite is quoted in the literature in two different units, which is a useful accident: 17.4 megapascals per root millimetre in Pickering's form for air-cooled mild steels, and 600 megapascals per root micrometre elsewhere [10]. Converted to a common basis these are 0.550 and 0.600 megapascals per root metre — two independently reported figures agreeing to nine per cent, which is about as much confidence as a materials constant ever earns.

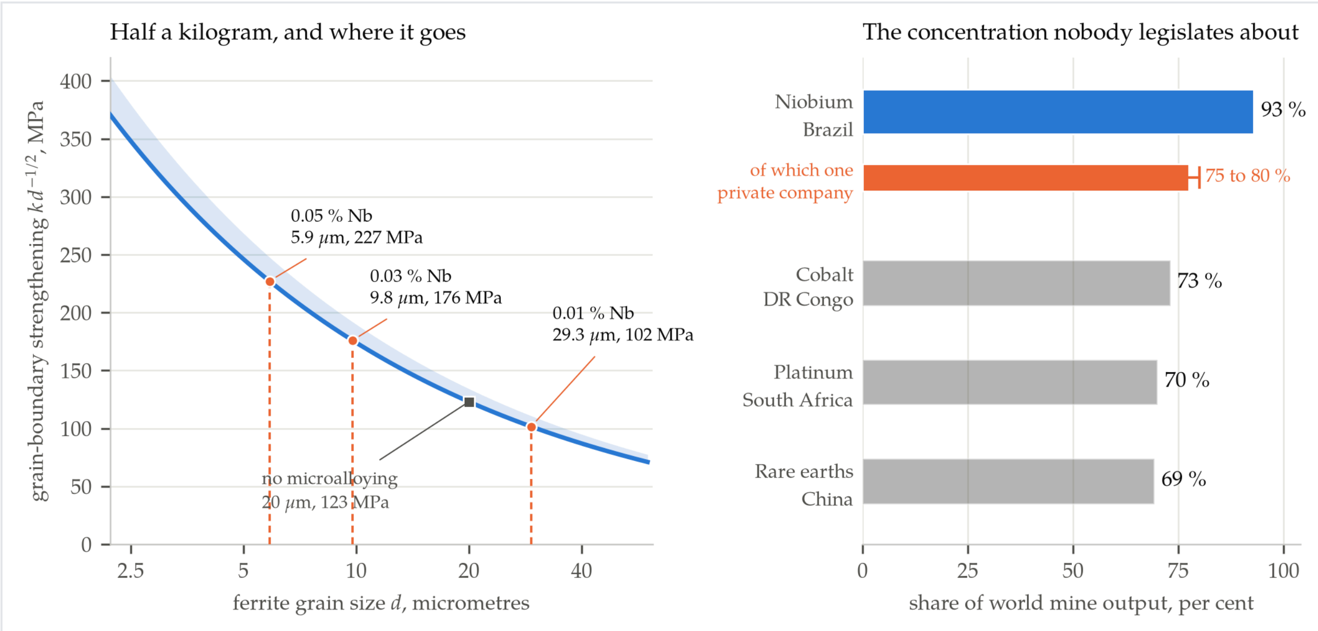


**FIGURE** Left: the grain-boundary strengthening available from refining ferrite, against the grain size that a given niobium addition can actually hold. The band is the disagreement between the two reported forms of the Hall-Petch constant; the markers are the Smith-Zener limit for 2.5-nanometre carbides at three addition levels, with the volume fraction computed from the addition rather than assumed. Right: leading-producer share of world mine output for four commodities, with the share attributed to a single company shown separately for niobium.

On the lower constant, a 20-micrometre ferrite grain — an ordinary hot-rolled plate with nothing added — contributes 123 megapascals of yield strength. At 5.9 micrometres it contributes 227. Half a kilogram of niobium in a tonne of steel is worth about 104 megapascals, and it costs nothing in ductility or weldability, which is the part that makes it irreplaceable rather than merely useful.

## The Break-Even Steel Price

Now put a price on it. The pressure a pipe can hold relates to its wall thickness by Barlow's formula,

$$t = \frac{p\,D}{2\,S\,F},$$

so for a fixed pressure, diameter and design factor the wall thickness goes inversely as the specified minimum yield strength $S$. The API 5L grade ladder is therefore also a mass ladder [11].

| Grade | Minimum yield, MPa | Wall relative to X52 | Steel saved |
|---|---:|---:|---:|
| X52 | 359 | 1.000 | — |
| X60 | 415 | 0.865 | 13.5 % |
| X70 | 485 | 0.740 | 26.0 % |
| X80 | 555 | 0.647 | 35.3 % |

A tonne of X70 line pipe does the work of 1.351 tonnes of X52, so every tonne delivered saves 0.351 tonnes of steel. The niobium that makes it possible costs, at the average unit value of United States ferroniobium imports in 2024 and 2025 of 26 dollars per kilogram of contained niobium, about 13 dollars a tonne of steel [1], [2].

Those two numbers set a break-even. For the additive to cost as much as the steel it displaces, steel would have to be worth 13 dollars divided by 0.351 tonnes — about 37 dollars a tonne. Steel has not been near that price since before the First World War. The leverage is not marginal, and it is not a matter of judgement: it is an order of magnitude with two decimal places to spare.

The same arithmetic gives a way to check the tonnages, which turns out to be necessary. World mine output of 112,000 tonnes of contained niobium against 1,882.6 million tonnes of crude steel is 59.5 grams of niobium per tonne of steel [2], [12]. The figure quoted in the trade literature for global intensity is 55 to 60 grams a tonne, with about 100 for the advanced economies and 40 for China [4]. The United States checks the same way and independently: 8,400 tonnes of apparent consumption against 81.4 million tonnes of crude steel is 103 grams a tonne [1]. Two ratios derived from four agency figures, landing on numbers reported by people who were not computing ratios.

That matters because the tonnage itself is unstable. The same series put world output at 83,000 tonnes for 2023 and 112,000 for 2024 — a jump of a third in one year [1], [2]. The intensity check favours the higher figure; a different check, below, favours the lower. Either way the market is worth somewhere between 2.2 and 2.9 billion dollars a year at the ferroniobium unit value, and that number is the hinge of everything that follows.

## A Market Too Small to Enter

Under three billion dollars a year is very little. The world's copper mines ship 23 million tonnes a year against niobium's hundred thousand or so — a factor of 205 by mass — and a single large copper mine turns over more than the entire niobium industry [13].

Consider what entering it looks like. NioCorp's Elk Creek deposit in Nebraska is the most advanced niobium project outside Brazil and Canada. Its 2017 feasibility study put annualised output at 7,055 tonnes of ferroniobium, which at 65 per cent is 4,586 tonnes of contained niobium, or 4.1 per cent of world supply [14]. Initial capital cost was 1.14 billion dollars in the 2022 assessment and 1.85 billion in the revised one, a 62 per cent increase [15]. So a project that would add four per cent to world supply requires capital equal to about two-thirds of the whole market's annual revenue — and the study assumes a realised price of 39.60 dollars per kilogram of contained niobium, 52 per cent above what ferroniobium has actually been fetching.

Set against that: reserves of more than 21 million tonnes, 14 million of them in Brazil, which at current rates is 188 years of world supply and 135 years for Brazil alone [2]. An incumbent with a century of reserves and the ability to flex output faces no credible entry threat, because the threat would have to be financed against a price the incumbent sets.

> **IF THE CUSTOMERS WOULD NOT NOTICE A TENFOLD RISE IN THE PRICE, WHY IS** the price not ten times higher?

Because the demand that matters is not the demand of existing users. A steelmaker already rolling X70 would indeed absorb a doubling of ferroniobium without blinking; 13 dollars a tonne becomes 26 against a product worth several hundred. But the marginal buyer is not that mill. It is a mill deciding whether to install the thermomechanical rolling control, the temperature instrumentation and the requalification programme that microalloying requires in the first place — a decision taken once, on a view of costs over decades. Price that decision badly and the mill goes on making thicker steel, or switches its alloy design to vanadium, and it does not come back. The ceiling on a niobium price is set by non-adoption, not by substitution among adopters, which is why the dominant producer behaves like a market developer — funding research, taking equity in downstream ventures, flexing volume to damp price swings — rather than like a monopolist collecting rent [16].

The direction of substitution in 2018 shows the strategy working. China's GB/T 1499.2-2018 rebar standard, in force that November, abolished the 335 megapascal grade and pushed the market to 400 megapascals and above [17]. Ferrovanadium, the main alternative microalloying route, went from about 26 dollars a kilogram of vanadium in mid-2017 to a peak near 140 in October 2018 — more than five times the price of ferroniobium [18]. Steelmakers moved toward niobium, not away from it. A producer that had spent the previous decade extracting the maximum rent would have had nothing cheap to sell into that window.

## The Customers Bought the Shop

If entry is irrational and the incumbent will not raise its price, what does a worried buyer actually do? In 2011 the world's largest steelmakers answered the question directly: they bought the monopolist.

In September of that year a consortium assembled by CITIC Metal with Baosteel, Anshan, Shougang and Taiyuan Iron and Steel paid 1.95 billion dollars for 15 per cent of the Araxá producer. A second consortium — Nippon Steel, JFE, the trading house Sojitz and Japan's state resources agency, with POSCO and the Korean national pension fund, each of the six holding 2.5 per cent — paid 1.8 billion for another 15 per cent [19], [20], [21]. The two tranches imply enterprise values of 13.0 and 12.0 billion dollars, agreeing to 8 per cent, and together they moved 3.75 billion dollars for 30 per cent of a company whose entire world market turns over less than three billion a year. Valued against the revenue of the product it sells, that is a multiple of about four and a half.

These were not financial investors hunting a return. They were the customers. The logic is exactly the logic of the previous section read backwards: if you cannot build a competitor for less than the market is worth, and the incumbent's pricing is already the thing protecting your margin, then the rational hedge is a seat at the table rather than a mine of your own. Every other niobium asset of consequence has changed hands the same way, to buyers wanting a position rather than a fight — Magris Resources bought the Niobec mine in Quebec from IAMGOLD in January 2015 for 530 million dollars, and China Molybdenum bought Anglo American's Brazilian niobium and phosphate business in 2016 for 1.7 billion [22], [23].

Brazilian politics has never quite accepted the smallness of the number. Niobium has been a recurring theme of resource nationalism there, described in a 2016 campaign visit to Araxá as potentially more important than petroleum, and the country's reserve position — two-thirds of the world's — makes the rhetoric easy [24]. The mineral royalty compounds the impression of value forgone, since its base is the producer's costs before transformation rather than the value of the ferroniobium sold [25]. But reserves are not revenue. A resource that will last two centuries at a turnover of under three billion dollars is not an oil industry; it is a specialty chemicals business that happens to sit on a hill.

## Where the Concentration Actually Bites

None of which means the concentration is harmless. It means the harm is not where the tonnage is.

| | Microalloyed steel | Specialty grades |
|---|---|---|
| Share of niobium demand | about 90 per cent | a few per cent |
| Substitutes in service | vanadium, molybdenum, titanium | none |
| Cost of a year's interruption | thicker steel | grounded programmes |
| Buffer that exists | mill and trader stocks | national stockpiles |

The specialty end is very small and completely inelastic. All commercial superconductor and nuclear-magnetic-resonance uses together consume of the order of 260 tonnes of niobium metal a year, which is 0.23 per cent of world supply [26]. The Large Hadron Collider's magnets contain 1,200 tonnes of cable holding 470 tonnes of niobium-titanium, so the alloy in one accelerator is roughly a year of the world's entire superconductor-grade niobium production. ITER's toroidal-field and central-solenoid magnets needed more than 600 tonnes of niobium-tin strand, a procurement that required world strand capacity to rise from about 15 tonnes a year to 100 [27]. Rocket nozzle extensions use C-103, which is niobium with 10 per cent hafnium and 1 per cent titanium, and there is no nickel superalloy that does the job at those temperatures.

For that fraction, substitution is not available at any price, which is what stockpiles are for. Niobium entered the American strategic inventory in 1943 as columbite ore and was designated a strategic and critical material in 1946; it has appeared on every critical-minerals list since 2018 [28]. In 2026 the Defense Logistics Agency went to market for 1,288,082 pounds — 584.3 tonnes, a conversion that checks to within 0.01 per cent — of vacuum-grade ferroniobium, under a solicitation with a ceiling of 160 million dollars [29]. That ceiling is about 5.5 per cent of one year of the world niobium market, which tells you both how affordable the insurance is and how narrow the exposure being insured against.

The second real exposure is informational, and it is the direct consequence of one private company holding the position. Nobody outside it knows the size of the market to better than a third. The producer's own reported 2023 sales were 92,000 tonnes of ferroniobium equivalent, which at 65 per cent is 59,800 tonnes of contained niobium [3]. That is 72 per cent of the 83,000-tonne world estimate for 2023 and 53 per cent of the 112,000-tonne estimate for 2024, and only the first of those is compatible with the 75-to-80 per cent share the company is independently credited with. Meanwhile the intensity check earlier in this piece favours the second. Both figures cannot be right, and no third party can adjudicate, because in a market with one dominant private seller the statistics are a disclosure rather than a measurement.

Which suggests the general lesson, and it is not the one the concentration tables imply. A supply share is not a risk measure. What matters is what a buyer would pay to avoid an interruption, and for 93 per cent of niobium's tonnage the answer is about 13 dollars a tonne of steel and a willingness to wait a quarter. For the sliver that goes into a magnet or a nozzle the answer is effectively unbounded and nothing substitutes — which is why 584 tonnes is a sensible thing for a government to buy and a hundred thousand would not be. And the reason the first number has stayed so small for seventy years is that the owner of the hill worked out early what its real competitor was. The alternative to selling niobium cheaply is not selling niobium dearly. It is not selling niobium.

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